

Alberto Andrés Valdés González.

Degree: Mathematical Engineer.

Work position: ML-Engineer.

Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com

Location: Santiago, Chile.

Differential inequation

Problem:

Let f be a function that satisfies the following conditions:

- $f \in C^1(a, b)$
- $\lim_{x \rightarrow a^+} f(x) = +\infty$
- $\lim_{x \rightarrow b^-} f(x) = -\infty$
- $f'(x) + f(x)^2 \geq -1 \quad \forall x \in (a, b)$

Prove that $(b - a) \geq \pi$ and give an example where $(b - a) = \pi$.

Solution:

We are going to start with:

$$f'(x) + [f(x)]^2 \geq -1$$

$\Rightarrow$

$$f'(x) \geq -(1 + [f(x)]^2)$$

$\Rightarrow$

$$\frac{df}{dx} \geq -(1 + [f(x)]^2)$$

$\Rightarrow$

$$\frac{df}{(1 + [f(x)]^2)} \geq -dx$$

We can consider $t, v : (t, v) \subset (a, b), t > a, u < b$

$\Rightarrow$

Remember that integration preserves inequality unlike differentiation.

$$\int_t^v \frac{df}{(1 + [f(x)]^2)} \geq - \int_t^v dx$$

$\Rightarrow$

$$\int_{f(t)}^{f(v)} \frac{u}{(1 + [u]^2)} \geq -(v - t)$$

$\Rightarrow$

$$\arctan [f(v)] - \arctan [f(t)] \geq -(v - t)$$

$\Rightarrow$

We can apply limits

$$\lim_{v \rightarrow b^+} (\arctan [f(v)] - \arctan [f(t)]) \geq - \lim_{v \rightarrow b^+} (v - t)$$

$\Rightarrow$

$$\left(\lim_{v \rightarrow b^+} \arctan [f(v)] \right) - \arctan [f(t)] \geq -(b - t)$$

$\Rightarrow$

$$\left(\lim_{v \rightarrow b^+} \arctan [f(v)] \right) - \arctan [f(t)] \geq -(b - t)$$

$\Rightarrow$

$$-\frac{\pi}{2} - \arctan [f(t)] \geq -(b - t)$$

We can apply limits again

$$-\frac{\pi}{2} - \lim_{t \rightarrow a^-} (\arctan [f(t)]) \geq - \lim_{t \rightarrow a^-} (b - t)$$

$\Rightarrow$

$$-\frac{\pi}{2} - \frac{\pi}{2} \geq -(b - a)$$

$\Rightarrow$

$$-\pi \geq -(b - a)$$

$\Rightarrow$

$$(b - a) \geq \pi$$

If we consider $f(x) = -\tan(x)$, $a = -\frac{\pi}{2}$, $b = \frac{\pi}{2}$ we have the equality.
